

DOYLESTOWN, PA, USA

WMO: 725113

Lat: 40.330N

Lon: 75.123W

Elev: 120

StdP: 99.89

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54786

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.2	-21.0	0.6	-11.1	-18.4	0.7	-8.6	10.3	-2.0	9.2	-2.9	3.4	320	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.9	23.3	31.5	22.6	30.0	21.9	24.8	30.1	24.1	28.8	23.4	27.7	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.3	26.7	22.7	17.6	26.0	22.2	17.1	25.5	75.8	30.1	72.7	28.9	70.0	27.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.2	-15.3	35.3	2.7	1.9	-17.2	36.6	-18.8	37.8	-20.4	38.8	-22.4	40.2		
(5)			-16.1	26.3	2.7	1.2	-18.1	27.2	-19.6	28.0	-21.1	28.7	-23.1	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-0.9	0.4	4.8	11.2	16.5	21.3	24.0	23.1	19.5	13.0	7.1	2.1		
	DBStd	9.74	5.72	5.18	5.20	4.90	4.24	3.46	2.69	2.84	3.62	4.52	4.64	4.91		
	HDD10.0	1212	341	271	177	41	2	0	0	0	21	110	250			
	HDD18.3	2888	597	502	420	222	89	13	1	2	29	174	336	503		
	CDD10.0	1902	2	3	14	77	202	338	435	405	283	114	24	5		
	CDD18.3	535	0	0	0	7	31	101	177	148	63	9	0	0		
	CDH23.3	4528	0	0	3	87	304	869	1566	1183	458	58	1	0		
	CDH26.7	1514	0	0	0	24	82	277	598	394	129	9	0	0		
(14) Wind	WSAvg	2.4	2.9	3.0	3.0	2.8	2.1	2.0	1.8	1.6	1.8	2.2	2.5	2.7		
(15) Precipitation	PrecAvg	1210	92	70	105	98	93	108	119	107	112	104	87	109		
	PrecMax	1671	167	159	197	259	172	244	259	329	278	279	145	258		
	PrecMin	882	43	21	27	33	24	24	22	34	13	8	29	26		
	PrecStd	192	35	32	44	49	37	56	47	63	70	68	32	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.7	22.7	29.6	31.4	33.3	35.3	34.1	32.5	27.7	21.6	18.0		
		MCWB	13.7	13.7	14.8	17.9	21.4	23.4	24.2	23.7	22.7	20.4	16.1	14.9		
	2%	DB	13.5	14.4	18.8	25.9	29.0	31.6	33.1	32.3	29.9	24.8	19.0	14.8		
		MCWB	11.2	10.3	12.7	16.6	20.5	22.6	23.4	23.4	21.5	19.0	15.0	12.9		
	5%	DB	9.8	11.0	16.0	22.7	27.0	29.8	31.6	30.4	27.9	22.6	17.0	12.3		
		MCWB	6.9	6.9	10.4	14.9	19.1	21.6	22.8	22.5	20.3	17.8	12.9	10.0		
	10%	DB	7.2	7.7	13.0	20.0	24.5	28.0	30.0	28.8	26.1	20.4	14.4	9.4		
		MCWB	4.9	4.4	8.9	13.4	18.1	20.8	22.0	21.7	19.7	16.4	11.3	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.2	16.2	20.5	23.0	24.6	25.9	26.1	24.7	22.2	17.9	16.3		
		MCDB	16.4	18.1	21.4	26.5	28.9	30.6	32.6	31.6	29.9	25.1	20.0	17.2		
	2%	WB	11.3	11.2	13.9	18.2	21.7	23.8	24.9	24.8	23.4	20.6	16.0	13.2		
		MCDB	12.9	14.3	17.1	23.5	27.1	29.1	30.5	29.4	26.8	23.7	18.1	14.3		
	5%	WB	7.7	7.1	11.8	16.3	20.5	23.0	24.1	23.9	22.4	19.0	13.9	10.5		
		MCDB	9.4	9.6	14.8	21.2	25.4	27.9	29.0	28.2	25.4	21.7	15.9	12.0		
	10%	WB	5.1	4.8	9.2	14.5	19.3	22.0	23.4	23.1	21.4	17.2	12.1	7.5		
		MCDB	7.1	7.5	13.0	19.0	23.3	26.6	27.9	26.8	24.3	19.4	13.8	8.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	9.8	11.1	10.9	10.5	10.5	10.1	10.4	9.9	9.4	7.7		
		MCDBR	11.6	12.9	13.7	14.8	13.5	12.4	12.2	11.6	12.6	12.2	11.8	10.5		
	5% WB	MCWBR	9.5	9.7	9.2	7.8	6.5	5.3	4.3	4.3	5.4	6.7	8.0	9.2		
		MCWBR	10.6	11.8	12.8	12.8	12.1	10.9	10.3	9.4	9.5	9.8	10.0	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.319	0.356	0.386	0.432	0.490	0.489	0.483	0.397	0.356	0.323	0.305		
	taud		2.458	2.405	2.374	2.319	2.219	2.094	2.115	2.118	2.328	2.455	2.509	2.512		
	Ebn at Noon		859	897	895	887	850	798	795	787	840	843	832	833		
	Edh at Noon		78	95	110	124	140	159	155	150	114	90	75	69		
(44) All-Sky Solar Radiation	RadAvg		1.87	2.78	3.67	4.76	5.30	5.80	5.94	5.13	4.27	3.01	2.13	1.61		
	RadStd		0.16	0.30	0.29	0.44	0.61	0.49	0.29	0.33	0.50	0.32	0.19	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+68

Nomenclature: See separate page